

Architectural Evolution of Stable Diffusion: A Technical Analysis of Stable Diffusion 3.5 and Comparison with FLUX

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Abstract

The field of generative image synthesis has undergone rapid development over the past several years, evolving from convolutional latent diffusion architectures to large-scale transformer-based generative models. Stable Diffusion 3.5 (SD3.5) represents the most mature member of the Stable Diffusion family and introduces significant architectural improvements over previous generations, including Stable Diffusion 2 (SD2), Stable Diffusion XL (SDXL), and Stable Diffusion 3 (SD3). In parallel, FLUX introduces a distinct generative paradigm based on Rectified Flow Transformers and advanced spatial reasoning mechanisms.

This report provides a technical analysis of the architectural evolution of Stable Diffusion, examines the major innovations introduced in SD3.5, and presents a detailed comparison between SD3.5 and FLUX.

1 Introduction

Diffusion models have become the dominant paradigm for high-fidelity image generation. Since the release of Stable Diffusion in 2022, the architecture has evolved through multiple generations, each addressing limitations of previous designs.

The progression from SD2 to SD3.5 reflects a broader trend within machine learning: the replacement of convolutional architectures with transformers capable of modeling long-range dependencies and complex multimodal interactions.

Unlike earlier generations that relied primarily on U-Net architectures, SD3 and SD3.5 adopt Multi-Modal Diffusion Transformers (MMDiT), enabling direct interaction between image and text tokens. Simultaneously, FLUX introduces an alternative approach based on Rectified Flow Transformers, offering improvements in spatial reasoning and generation consistency.

Rather than representing a strict linear progression, SD3.5 and FLUX should be viewed as two parallel state-of-the-art transformer-based generative architectures that prioritize different design objectives.

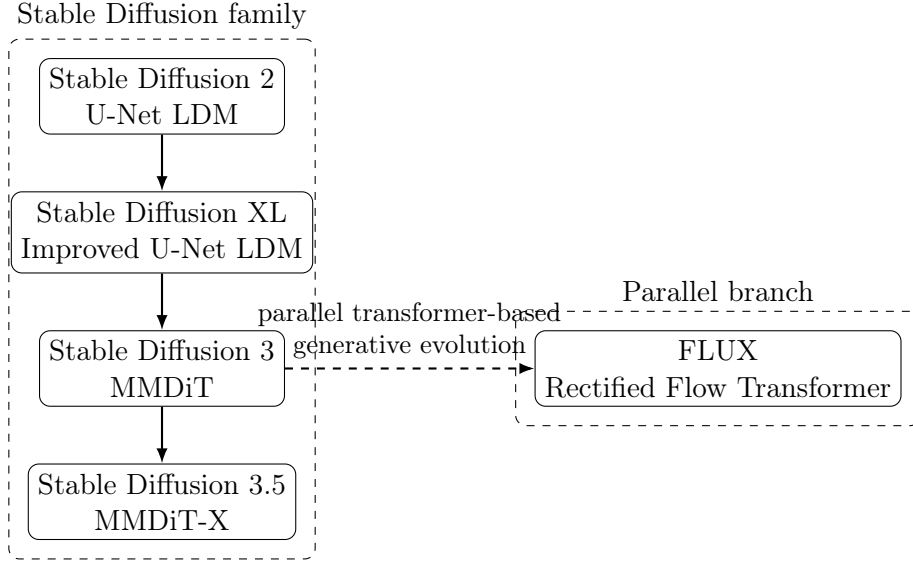


Figure 1: Stable Diffusion and FLUX are related but not strictly sequential model families. SD3.5 is the mature continuation of the Stable Diffusion line, while FLUX is a parallel transformer-based architecture.

2 Evolution of Stable Diffusion

2.1 Stable Diffusion 2

Stable Diffusion 2 is based on the Latent Diffusion Model (LDM) framework. The architecture consists of three primary components:

- Variational Autoencoder (VAE)
- OpenCLIP text encoder
- U-Net denoising network

The VAE compresses images into latent representations, allowing diffusion to occur in a lower-dimensional space and significantly reducing computational requirements.

Despite its success, SD2 exhibits several limitations:

- Weak compositional reasoning
- Limited global context understanding
- Poor attribute binding
- Difficulty handling complex scenes
- Limited prompt adherence

2.2 Stable Diffusion XL

Stable Diffusion XL introduced several major improvements while retaining the U-Net backbone.

2.2.1 Dual Text Encoder Design

SDXL employs two text encoders instead of one, providing richer semantic representations.

Benefits include:

- Improved prompt understanding
- Better attribute consistency
- Enhanced compositional reasoning
- Stronger text-image alignment

2.2.2 Improved Training Data

SDXL was trained on significantly larger and cleaner datasets, leading to:

- Better realism
- Sharper textures
- Higher image fidelity
- Improved object consistency

2.2.3 Enhanced Latent Representation

The latent representation captures more visual information than previous generations, producing images with greater detail and stability.

2.3 Stable Diffusion 3

Stable Diffusion 3 introduced the largest architectural shift in the history of Stable Diffusion.

The traditional U-Net architecture was replaced with:

MMDiT (Multi-Modal Diffusion Transformer)

Advantages include:

- Global attention mechanisms
- Better multimodal interaction
- Improved object relationships
- Stronger prompt adherence
- Better compositional reasoning

2.4 Stable Diffusion 3.5

Stable Diffusion 3.5 refines the concepts introduced in SD3 and improves both generation quality and training stability.

The model introduces:

- MMDiT-X architecture
- Improved attention routing
- Better token utilization
- Stronger semantic alignment
- Enhanced human generation
- More stable fine-tuning

3 Architectural Improvements in Stable Diffusion 3.5

3.1 MMDiT-X Architecture: Motivation and Design Evolution

3.1.1 Limitations of MMDiT in SD3

While MMDiT introduced a substantial improvement over U-Net by enabling direct interaction between image and text tokens through joint attention, several architectural limitations became apparent during SD3 training and inference.

The original MMDiT design applies the *same attention weight matrices* to both image and text token streams. Formally, for a joint token sequence $z = [z_{\text{img}}; z_{\text{txt}}]$, the attention computation is:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \quad (1)$$

where Q, K, V are computed from the concatenated joint token sequence using shared projection matrices.

This design imposes a fundamental constraint: the same learned attention routing is applied regardless of whether the token originates from the image stream or the text stream. The two modalities have very different statistical distributions and semantic roles, yet SD3 does not allow the model to independently specialize routing within each stream.

Observed limitations of MMDiT in SD3 include:

- **Coupled attention routing.** Image tokens and text tokens compete within the same attention head space, limiting the degree to which each stream can independently specialize.
- **Semantic leakage.** Text conditioning can inadvertently interfere with image structure, particularly in fine-tuning scenarios where the text encoder embeddings drift.
- **Suboptimal token mixing at scale.** As model depth increases, the fixed joint attention pattern without explicit routing mechanisms becomes a scalability bottleneck.
- **Unstable fine-tuning behavior.** The coupling between streams makes it difficult to selectively adapt the image pathway without disturbing the text pathway during parameter-efficient fine-tuning.

3.1.2 Key Innovations in MMDiT-X

MMDiT-X addresses these limitations through several targeted architectural modifications.

Independent Weight Streams per Modality. The central innovation of MMDiT-X is the introduction of *independent projection matrices* for the image and text token streams within each transformer block.

Under MMDiT (SD3), the projection matrices W_Q , W_K , W_V are applied to the concatenated joint sequence. Under MMDiT-X (SD3.5), dedicated projection matrices are maintained for each modality:

$$Q_{\text{img}} = W_Q^{\text{img}} z_{\text{img}}, \quad Q_{\text{txt}} = W_Q^{\text{txt}} z_{\text{txt}} \quad (2)$$

$$K_{\text{img}} = W_K^{\text{img}} z_{\text{img}}, \quad K_{\text{txt}} = W_K^{\text{txt}} z_{\text{txt}} \quad (3)$$

The joint attention is then computed over the concatenated keys and queries:

$$Q = [Q_{\text{img}}; Q_{\text{txt}}], \quad K = [K_{\text{img}}; K_{\text{txt}}] \quad (4)$$

This allows the model to learn different inductive biases for each modality while retaining full cross-modal attention. The image stream can specialize in spatial structure and texture statistics, while the text stream develops representations optimized for semantic conditioning.

Extended Attention Routing. The “X” in MMDiT-X refers to the *extended* attention routing mechanism. In SD3, attention heads are allocated uniformly across all transformer blocks at every depth level. MMDiT-X introduces a non-uniform routing scheme, allocating more expressive attention capacity to deeper transformer blocks where semantic reasoning occurs, and more spatially-sensitive processing to shallower blocks where low-level structure is established.

This mirrors findings in large language model research, where later transformer layers are responsible for higher-order semantic composition while earlier layers encode syntactic and positional structure.

Improved AdaLN Conditioning. Both MMDiT and MMDiT-X use Adaptive Layer Normalization (AdaLN) to inject timestep and text conditioning into each transformer block. In SD3, the AdaLN conditioning vector is computed from a single global pooled embedding. MMDiT-X improves this by computing *separate* AdaLN scale and shift parameters for the image pathway and the text pathway:

$$\hat{z}_{\text{img}} = \gamma_{\text{img}}(c) \text{LN}(z_{\text{img}}) + \beta_{\text{img}}(c) \quad (5)$$

$$\hat{z}_{\text{txt}} = \gamma_{\text{txt}}(c) \text{LN}(z_{\text{txt}}) + \beta_{\text{txt}}(c) \quad (6)$$

where c represents the combined conditioning signal (timestep embedding + pooled text embedding) and γ , β are learned affine parameters. This provides each stream with independent normalization dynamics, reducing the cross-stream interference observed in SD3.

3.1.3 Architectural Comparison: MMDiT vs. MMDiT-X

Design Property	MMDiT (SD3)	MMDiT-X (SD3.5)
Attention projection matrices	Shared	Per-modality
Attention routing	Uniform	Extended (non-uniform)
AdaLN conditioning	Single pathway	Dual pathway
Stream independence	Partial	Full
Fine-tuning stability	Moderate	High
Prompt-image decoupling	Limited	Improved
Semantic alignment	High	Very High

Table 1: Architectural differences between MMDiT (SD3) and MMDiT-X (SD3.5)

3.1.4 Why These Changes Matter in Practice

The architectural separation of image and text pathways in MMDiT-X has direct consequences for generation quality.

In SD3, because both modalities share projection matrices, the model must learn a single set of attention weights that simultaneously serves both visual structure and language semantics. This is analogous to forcing the same neural pathway to specialize in two fundamentally different representation spaces simultaneously.

MMDiT-X removes this constraint. The image pathway develops strong spatial and textural inductive biases. The text pathway develops strong semantic and relational inductive biases. Cross-modal interaction still occurs through the joint attention mechanism, but each modality contributes from a more specialized representation.

For tasks relevant to dense urban scene generation, such as pedestrian augmentation, this translates directly into:

- **Better attribute binding.** When a prompt specifies “a pedestrian wearing a red jacket near a bus stop,” MMDiT-X more reliably binds “red” to “jacket” rather than to unrelated scene elements.
- **Improved spatial consistency.** The independent image pathway better preserves the geometric structure of the background scene during compositing.
- **More stable LoRA fine-tuning.** Because image and text pathways are decoupled, LoRA adapters applied to the image stream do not inadvertently interfere with text conditioning, enabling more targeted domain adaptation.
- **Reduced prompt drift at high guidance scales.** The extended attention routing provides more stable conditioning at classifier-free guidance values above 7.0, which are commonly used for high-fidelity generation.

3.2 Improved Prompt Adherence

Compared with SD3, SD3.5 demonstrates:

- Better attribute binding
- Reduced prompt drift
- Improved instruction following
- Better handling of long prompts

3.3 Improved Human Generation

Compared with SD3, SD3.5 exhibits:

- Better body proportions
- Improved hand generation
- Better facial consistency
- Improved pose preservation
- More realistic anatomy

3.4 Small Object Representation

Examples include:

- Pedestrians
- Traffic lights
- Road signs
- Distant vehicles
- Urban infrastructure

3.5 Fine-Tuning Stability

Benefits include:

- Better convergence
- Reduced instability
- Improved reproducibility
- Lower artifact generation

4 Motivation Behind Architectural Changes in Stable Diffusion 3.5

Stable Diffusion 3.5 did not emerge as a completely new architecture. Instead, it represents the result of several years of iterative improvements addressing the limitations observed in previous generations, including Stable Diffusion 2 (SD2), Stable Diffusion XL (SDXL), and Stable Diffusion 3 (SD3).

Each major architectural block in SD3.5 was introduced to solve a specific bottleneck that constrained image quality, prompt adherence, compositional reasoning, or training scalability.

4.1 Why VAE Remains Necessary

Despite the transition from convolutional architectures to transformers, SD3.5 continues to utilize a Variational Autoencoder (VAE).

The primary motivation is computational efficiency.

Consider a 1024×1024 RGB image:

$$1024 \times 1024 = 1,048,576 \tag{7}$$

pixels must be processed.

Transformer attention scales approximately as:

$$O(N^2) \tag{8}$$

where N represents the number of tokens.

Applying full transformer attention directly to pixel-space images would therefore be computationally prohibitive.

The VAE compresses images into latent representations before diffusion takes place, reducing memory requirements by more than an order of magnitude while preserving most semantic information.

Consequently, although SD3.5 introduces a completely different denoising architecture, the latent diffusion paradigm remains essential.

4.2 Why CLIP Alone Became Insufficient

Stable Diffusion 2 primarily relied on OpenCLIP embeddings for text conditioning.

While CLIP provides strong image-text alignment, it was not originally designed as a language reasoning model.

Several limitations became apparent:

- Weak understanding of long prompts.
- Poor compositional reasoning.
- Difficulty handling multiple attributes.
- Limited relational understanding.

For example, prompts such as

“A woman holding a red umbrella while standing behind a blue car.”

often resulted in attribute confusion or incorrect object relationships.

As prompt complexity increased, the limitations of CLIP became increasingly visible.

4.3 Why U-Net Became a Bottleneck

U-Net architectures were highly effective during the early era of latent diffusion.

However, convolution-based processing exhibits inherent limitations:

- Local receptive fields.
- Weak long-range interactions.
- Inefficient multimodal reasoning.
- Difficulty modeling complex object relationships.

Suppose a scene contains:

A pedestrian standing behind a vehicle.

The pedestrian and vehicle may occupy distant regions of the latent representation.

Under convolutional processing, information must propagate through multiple layers before interactions can be learned.

This becomes increasingly problematic for:

- Multi-object scenes.
- Counting tasks.
- Spatial relationships.
- Human-object interactions.

These limitations motivated the transition toward transformer-based architectures.

4.4 Why Flow Matching Replaced Traditional Noise Prediction

Previous Stable Diffusion models trained by predicting noise:

$$L_{\text{diff}} = \mathbb{E} [\|\epsilon - \epsilon_{\theta}(x_t, t)\|^2]. \quad (9)$$

Although successful, noise prediction becomes increasingly difficult when scaling to large transformer architectures.

Flow Matching reformulates the problem as velocity prediction:

$$\frac{dx}{dt} = v_{\theta}(x_t, t). \quad (10)$$

Advantages include:

- Smoother optimization.
- Improved convergence.
- Better scaling behavior.
- More stable transformer training.

Flow Matching therefore became a natural choice for large multimodal transformer architectures.

4.5 Why RoPE Replaced Traditional Positional Encoding

Early diffusion architectures relied on sinusoidal positional encodings.

Although effective, these encodings were originally developed for sequence modeling rather than multimodal transformer architectures.

SD3 and SD3.5 adopt Rotary Position Embeddings (RoPE).

Advantages include:

- Better relative position representation.
- Improved long-range attention.
- Stronger transformer scalability.
- Efficient implementation.

Today, RoPE has become the dominant positional encoding strategy across modern large-scale transformer architectures.

5 Text Conditioning Pipeline in Stable Diffusion 3.5

One of the most important innovations introduced by SD3.5 is its multi-encoder text conditioning pipeline.

Unlike previous generations that relied primarily on CLIP, SD3.5 combines three separate language representations:

- CLIP-L/14
- CLIP-G/14
- T5-XXL

Each encoder contributes complementary information.

5.1 CLIP-L/14

CLIP-L/14 is responsible for strong image-text alignment.

Its primary strengths include:

- Object recognition.
- Visual concept representation.
- Image-language correspondence.

Examples:

- Person
- Car
- Bicycle
- Traffic light

CLIP-L excels at identifying visual concepts that commonly appear in image generation tasks.

5.2 CLIP-G/14

CLIP-G is a larger OpenCLIP model.

Compared with CLIP-L, it provides:

- Richer semantic representations.
- Improved context understanding.
- Better scene-level conditioning.

While CLIP-L focuses on object-level understanding, CLIP-G contributes broader scene-level information.

5.3 T5-XXL

T5-XXL is the largest language model within the SD3.5 conditioning pipeline.

Unlike CLIP, T5 was designed primarily for language understanding.

It excels at:

- Syntax.
- Grammar.
- Logical reasoning.
- Attribute relationships.
- Long-context understanding.

For example:

“A young woman wearing a yellow raincoat while holding a red umbrella.”

T5 understands:

- Subject.
- Action.
- Attribute assignment.
- Object relationships.

more effectively than CLIP.

5.4 Why Three Encoders Are Used Simultaneously

Each encoder specializes in different aspects of prompt understanding.

Encoder	Primary Strength	Limitation
CLIP-L	Visual alignment	Limited reasoning
CLIP-G	Rich semantics	Limited language structure
T5-XXL	Language reasoning	Weaker visual alignment

Table 2: Complementary strengths of SD3.5 text encoders

Combining all three allows SD3.5 to simultaneously achieve:

- Strong visual grounding.
- Rich semantic understanding.
- Improved language reasoning.
- Better prompt adherence.

5.5 Benchmark Evidence

The introduction of T5-XXL and multi-encoder conditioning significantly improved performance on prompt-following benchmarks.

Reported improvements include:

- Higher DPG-Bench scores.
- Better GenEval performance.
- Improved attribute binding.
- Stronger multi-object reasoning.

6 Denoising Architecture Evolution

6.1 U-Net in SD2 and SDXL

Earlier Stable Diffusion generations employed U-Net architectures for denoising.

The workflow can be summarized as:

Latent Representation $\rightarrow$ U-Net $\rightarrow$ Noise Prediction

The architecture relies primarily on convolutional operations.

6.2 Limitations of Convolutional Denoising

Convolutions operate locally.

Consequently, information propagation requires multiple layers before distant regions can interact.

This becomes problematic for:

- Object counting.
- Human-object interactions.
- Spatial reasoning.
- Multi-person scenes.

As scene complexity increases, U-Net becomes increasingly inefficient.

6.3 MMDiT in Stable Diffusion 3

SD3 replaces U-Net with MMDiT.

The architecture treats both image and text as tokens processed within a transformer.

Advantages include:

- Global attention.
- Better multimodal fusion.
- Improved compositional reasoning.
- Stronger prompt adherence.

6.4 MMDiT-X in Stable Diffusion 3.5

MMDiT-X further improves:

- Attention routing.
- Token mixing.
- Transformer scalability.
- Semantic alignment.

Compared with SD3, MMDiT-X exhibits more stable generation and stronger prompt understanding.

6.5 Joint Attention Mechanism

The most important innovation introduced by MMDiT is Joint Attention.

Unlike traditional cross-attention mechanisms:

Text $\rightarrow$ Cross Attention $\rightarrow$ Image

Joint Attention allows:

Text Tokens $\leftrightarrow$ Image Tokens

to interact directly.

This significantly improves object relationships and attribute binding.

6.6 How Attention Changes the Denoising Process

The transition from U-Net to MMDiT fundamentally changes how denoising occurs.

Under U-Net:

Latent $\rightarrow$ Convolution $\rightarrow$ Local Feature Extraction $\rightarrow$ Noise Prediction

Under MMDiT:

Image Tokens + Text Tokens $\rightarrow$ Global Attention $\rightarrow$ Joint Reasoning $\rightarrow$ Velocity Prediction

A token can directly attend to every other token in the scene.

For example:

“A woman standing behind a red car.”

U-Net must infer the relationship through multiple convolutional layers.

MMDiT learns:

- Woman $\leftrightarrow$ Car
- Woman $\leftrightarrow$ Behind
- Car $\leftrightarrow$ Behind

directly through attention.

This significantly improves:

- Spatial reasoning.
- Multi-object composition.
- Attribute consistency.
- Prompt adherence.

6.7 Benchmark Evidence

The transition from U-Net to MMDiT produced measurable improvements.

Benchmark	SDXL	SD3	SD3.5
GenEval	~ 0.55	~ 0.62	$\sim 0.65+$
DPG-Bench	Baseline	Higher	Highest
Human Preference	Baseline	Improved	Improved Further
Attribute Binding	Moderate	High	Very High
Multi-object Reasoning	Moderate	High	Very High

Table 3: Representative benchmark trends across Stable Diffusion generations

These results demonstrate that the introduction of MMDiT and MMDiT-X substantially improved the ability of diffusion models to understand complex prompts and generate coherent scenes.

7 Flow Matching versus Rectified Flow

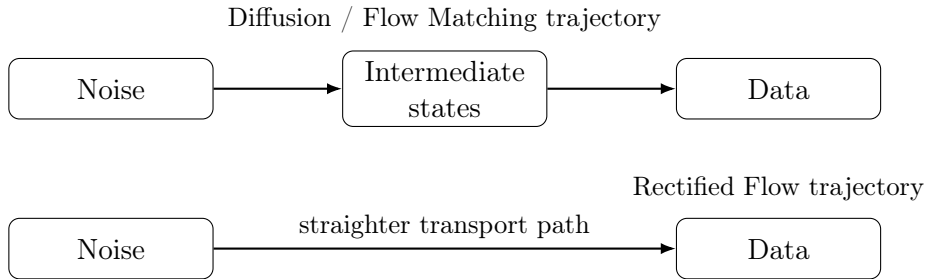


Figure 2: Conceptual comparison between conventional diffusion-style transport and Rectified Flow. Rectified Flow attempts to learn a straighter path from noise to data.

7.1 Traditional Diffusion

The conventional diffusion objective is

$$L_{\text{diff}} = \mathbb{E} [\|\epsilon - \epsilon_{\theta}(x_t, t)\|^2]. \quad (11)$$

7.2 Flow Matching

Stable Diffusion 3 and Stable Diffusion 3.5 adopt Flow Matching.

The velocity field is defined as

$$\frac{dx}{dt} = v_{\theta}(x_t, t). \quad (12)$$

The optimization objective becomes

$$L_{\text{FM}} = \mathbb{E} [\|v_t - v_{\theta}(x_t, t)\|^2]. \quad (13)$$

7.3 Rectified Flow

FLUX adopts Rectified Flow.

The corresponding objective is

$$L_{\text{RF}} = \mathbb{E} [\|(x_1 - x_0) - v_{\theta}(x_t, t)\|^2]. \quad (14)$$

7.4 Theoretical Comparison

Property	Flow Matching	Rectified Flow
Trajectory Complexity	Moderate	Low
Optimization Difficulty	Moderate	Lower
Sampling Efficiency	High	Very High
Stability	High	High
Transformer Compatibility	Excellent	Excellent

Table 4: Flow Matching versus Rectified Flow

8 MMDiT-X versus FLUX Transformer

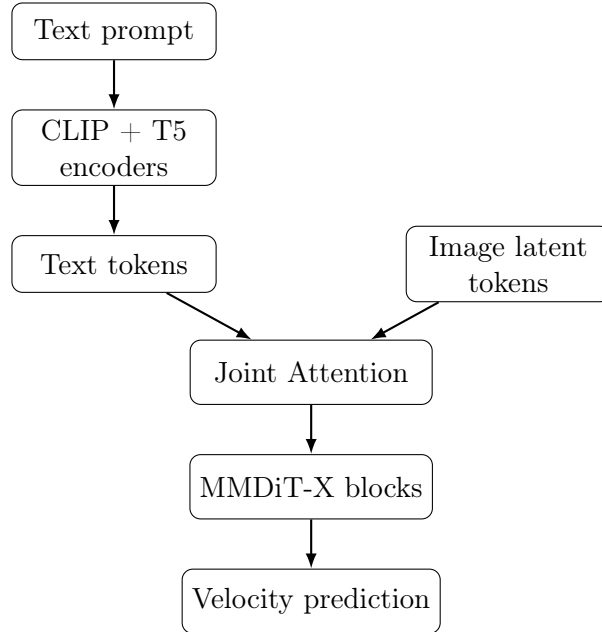


Figure 3: Simplified MMDiT-X pipeline in SD3.5. Text and image latent tokens interact through joint attention before velocity prediction.

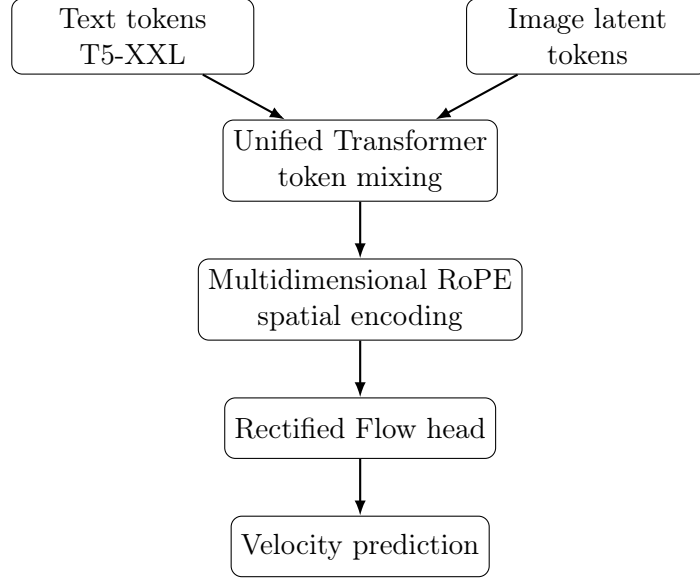


Figure 4: Simplified FLUX Transformer pipeline. FLUX emphasizes unified token mixing, multidimensional spatial encoding, and Rectified Flow velocity prediction.

8.1 Joint Attention in MMDiT-X

The defining innovation of MMDiT-X is Joint Attention.

Advantages include:

- Better multimodal fusion
- Improved prompt adherence
- Stronger semantic consistency

8.2 RoPE

Stable Diffusion 3.5 employs Rotary Position Embeddings (RoPE).

$$\text{RoPE}(x_n) = R(\theta_n) x_n \quad (15)$$

where

$$R(\theta_n) = \begin{bmatrix} \cos \theta_n & -\sin \theta_n \\ \sin \theta_n & \cos \theta_n \end{bmatrix}. \quad (16)$$

8.3 Multidimensional RoPE in FLUX

FLUX extends positional encoding into multidimensional spatial representations.

Advantages include:

- Better spatial awareness
- Stronger geometric consistency
- Improved perspective reasoning
- Better object placement

8.4 Spatial Reasoning

FLUX demonstrates stronger performance in:

- Perspective consistency
- Human scale consistency
- Multi-object relationships
- Scene geometry understanding

9 Stable Diffusion 3.5 versus FLUX

Feature	SD2	SDXL	SD3	SD3.5	FLUX
Backbone	U-Net	U-Net	MMDiT	MMDiT-X	RFT
Text Encoder	OpenCLIP	Dual CLIP	CLIP+T5	CLIP+T5	T5-XXL
Global Attention	Partial	Partial	Full	Full	Full
Flow Matching	No	No	Yes	Yes	RF
Position Encoding	Sin	Sin	RoPE	RoPE	Multi-RoPE
Human Generation	Medium	Good	Good	Very Good	Excellent
Multi-Person	Weak	Moderate	Good	Very Good	Excellent
Fine-Tuning	Excellent	Excellent	Good	Excellent	Moderate
VRAM Efficiency	Excellent	Good	Good	Good	Poor

Table 5: Evolution of Stable Diffusion and comparison with FLUX

10 Conclusion

Stable Diffusion 3.5 represents the most mature evolution of the Stable Diffusion family and introduces substantial improvements over SD2, SDXL, and SD3. Key innovations include MMDiT-X, improved prompt adherence, better human generation, stronger small-object representation, and more stable fine-tuning behavior.

FLUX represents a parallel evolution rather than a direct successor. Through Rectified Flow Transformers and multidimensional positional encoding, FLUX achieves superior spatial reasoning and stronger scene understanding. However, these advantages require significantly greater computational resources.

Consequently, SD3.5 and FLUX should be viewed as complementary state-of-the-art architectures optimized for different deployment scenarios.